

Algebraic Geometry 11

Flabby Sheaves, Exact Sequences, and Vanishing of Čech Cohomology

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1 Flabby Sheaves and Vanishing of Čech Cohomology

1.1 Problem 3

Let X be a topological space. A sheaf $\mathcal{F}$ on X is called *flabby* if for every inclusion of open sets $V \subseteq U$, the restriction map

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$$

is surjective.

We prove that if $\mathcal{U} = \{U_i\}_{i \in I}$ is an open covering of X and $\mathcal{F}$ is flabby, then

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \quad \text{for all } p > 0.$$

(a)

Assume that

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is an exact sequence of sheaves on X , and assume that $\mathcal{F}_1$ is flabby. We show that for every open set $U \subseteq X$, the map

$$\mathcal{F}_2(U) \longrightarrow \mathcal{F}_3(U)$$

is surjective.

Let $s \in \mathcal{F}_3(U)$. Since the sequence is exact as a sequence of sheaves, locally on U the section s lifts to sections of $\mathcal{F}_2$. Thus there exists an open covering $\{U_i\}$ of U and sections

$$t_i \in \mathcal{F}_2(U_i)$$

such that the image of t_i in $\mathcal{F}_3(U_i)$ is $s|_{U_i}$.

On overlaps $U_i \cap U_j$, we have

$$t_i|_{U_i \cap U_j} - t_j|_{U_i \cap U_j} \in \mathcal{F}_1(U_i \cap U_j).$$

We now glue the local liftings. Consider the set of pairs (V, t) , where $V \subseteq U$ is open and $t \in \mathcal{F}_2(V)$ lifts $s|_V$. Order these pairs by extension:

$$(V, t) \leq (W, t')$$

if $V \subseteq W$ and $t'|_V = t$.

By Zorn's Lemma, there exists a maximal such pair (V, t) . We claim that $V = U$.

Suppose not. Choose $x \in U \setminus V$. Since s is locally liftable, there exists an open neighbourhood W of x and a section

$$u \in \mathcal{F}_2(W)$$

lifting $s|_W$.

On $V \cap W$, both t and u lift the same section of $\mathcal{F}_3$. Therefore

$$t|_{V \cap W} - u|_{V \cap W} \in \mathcal{F}_1(V \cap W).$$

Since $\mathcal{F}_1$ is flabby, this section extends to a section

$$a \in \mathcal{F}_1(W).$$

Replace u by $u + a$. Then on $V \cap W$,

$$(u + a)|_{V \cap W} = t|_{V \cap W}.$$

Hence t and $u + a$ glue to a section of $\mathcal{F}_2(V \cup W)$ lifting $s|_{V \cup W}$, contradicting maximality.

Thus $V = U$, so s lifts globally. Therefore

$$\mathcal{F}_2(U) \longrightarrow \mathcal{F}_3(U)$$

is surjective.

(b)

Assume

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is exact, and suppose that $\mathcal{F}_1$ and $\mathcal{F}_2$ are flabby. We prove that $\mathcal{F}_3$ is flabby.

Let $V \subseteq U$ be open. We must prove that

$$\mathcal{F}_3(U) \longrightarrow \mathcal{F}_3(V)$$

is surjective.

Take $s \in \mathcal{F}_3(V)$. By part (a), because $\mathcal{F}_1$ is flabby, the map

$$\mathcal{F}_2(V) \longrightarrow \mathcal{F}_3(V)$$

is surjective. Hence there exists

$$t \in \mathcal{F}_2(V)$$

mapping to s .

Since $\mathcal{F}_2$ is flabby, t extends to a section

$$\tilde{t} \in \mathcal{F}_2(U).$$

Let $\tilde{s}$ be the image of $\tilde{t}$ in $\mathcal{F}_3(U)$. Then

$$\tilde{s}|_V = s.$$

Therefore the restriction map

$$\mathcal{F}_3(U) \longrightarrow \mathcal{F}_3(V)$$

is surjective, so $\mathcal{F}_3$ is flabby.

(c)

Let

$$0 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F}_1 \longrightarrow \cdots \longrightarrow \mathcal{F}_{p-1} \longrightarrow \mathcal{F}_p \longrightarrow \mathcal{F}_{p+1} \longrightarrow \cdots$$

be a long exact sequence of flabby sheaves.

We prove that for every open set $U \subseteq X$,

$$0 \longrightarrow \mathcal{F}_0(U) \longrightarrow \mathcal{F}_1(U) \longrightarrow \cdots \longrightarrow \mathcal{F}_{p-1}(U) \longrightarrow \mathcal{F}_p(U) \longrightarrow \mathcal{F}_{p+1}(U) \longrightarrow \cdots$$

is exact.

Exactness is automatic except possibly at the level of surjectivity onto kernels. Let

$$Z_p = \ker(\mathcal{F}_p \rightarrow \mathcal{F}_{p+1}).$$

Since the original sequence is exact as a sequence of sheaves, we have short exact sequences

$$0 \longrightarrow Z_p \longrightarrow \mathcal{F}_p \longrightarrow Z_{p+1} \longrightarrow 0.$$

We prove by induction that each Z_p is flabby. Since $\mathcal{F}_0$ is flabby, Z_1 is a quotient of flabby sheaves in the sequence

$$0 \longrightarrow Z_0 \longrightarrow \mathcal{F}_0 \longrightarrow Z_1 \longrightarrow 0.$$

Using part (b), the flabbiness propagates from one kernel sheaf to the next.

Now apply part (a) to each short exact sequence

$$0 \longrightarrow Z_p \longrightarrow \mathcal{F}_p \longrightarrow Z_{p+1} \longrightarrow 0.$$

Since Z_p is flabby, the map

$$\mathcal{F}_p(U) \longrightarrow Z_{p+1}(U)$$

is surjective.

But $Z_{p+1}(U)$ is precisely the kernel of

$$\mathcal{F}_{p+1}(U) \longrightarrow \mathcal{F}_{p+2}(U).$$

Therefore the sequence of global sections over U is exact.

(d)

Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open covering of X . Define

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p \in I} \mathcal{F}|_{U_{i_0} \cap \cdots \cap U_{i_p}}.$$

Then

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \Gamma(X, \mathcal{C}^p(\mathcal{U}, \mathcal{F})).$$

The boundary maps

$$d^p : \mathcal{C}^p(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^{p+1}(\mathcal{U}, \mathcal{F})$$

are defined by the usual alternating formula:

$$(d^p s)_{i_0, \dots, i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k s_{i_0, \dots, \widehat{i_k}, \dots, i_{p+1}} \Big|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}}.$$

We must show that

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \dots$$

is exact.

It is enough to check exactness on stalks. Let $x \in X$. Choose an index $i \in I$ such that $x \in U_i$.

On the stalk at x , the complex becomes the augmented Čech complex of the covering of a point. This complex is exact because it has a contracting homotopy.

Explicitly, define

$$h : C_x^p \longrightarrow C_x^{p-1}$$

by

$$(hs)_{i_0, \dots, i_{p-1}} = s_{i, i_0, \dots, i_{p-1}}.$$

Then one checks that

$$dh + hd = \text{id}.$$

Therefore the stalk complex is exact at every point $x \in X$.

Hence the sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \dots$$

is exact.

(e)

Assume now that $\mathcal{F}$ is flabby. We prove that each

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F})$$

is flabby.

For every multi-index $(i_0, \dots, i_p)$, the sheaf

$$\mathcal{F}|_{U_{i_0} \cap \dots \cap U_{i_p}}$$

is flabby. Indeed, if $V \subseteq W$ are open in X , then

$$\mathcal{F}(W \cap U_{i_0} \cap \dots \cap U_{i_p}) \longrightarrow \mathcal{F}(V \cap U_{i_0} \cap \dots \cap U_{i_p})$$

is surjective because $\mathcal{F}$ is flabby.

A product of flabby sheaves is flabby. Therefore

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} \mathcal{F}|_{U_{i_0} \cap \dots \cap U_{i_p}}$$

is flabby.

By part (d), we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \cdots$$

Since all the sheaves $\mathcal{C}^p(\mathcal{U}, \mathcal{F})$ are flabby, part (c) implies that taking global sections preserves exactness. Hence

$$0 \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow C^0(\mathcal{U}, \mathcal{F}) \longrightarrow C^1(\mathcal{U}, \mathcal{F}) \longrightarrow \cdots$$

is exact.

But the cohomology of this complex is precisely the Čech cohomology

$$\check{H}^p(\mathcal{U}, \mathcal{F}).$$

Therefore

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \quad \text{for all } p > 0.$$

1.2 Conclusion

For a flabby sheaf $\mathcal{F}$ on a topological space X , every section over an open subset extends to every larger open subset. This extension property forces the Čech complex associated to any open covering $\mathcal{U}$ to be exact in positive degrees. Consequently,

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \quad \text{for all } p > 0.$$

2 Theory: Flabby Sheaves and Vanishing of Čech Cohomology

A sheaf $\mathcal{F}$ on a topological space X is called *flabby* if for every inclusion of open sets $V \subseteq U$, the restriction map

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$$

is surjective.

Equivalently, every section over a smaller open set V extends to every larger open set U containing it. Thus flabbiness is a strong extension property.

Basic meaning of flabbiness

For an arbitrary sheaf, a section over an open set V does not necessarily extend to a larger open set U . In other words, the restriction map

$$\mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

need not be surjective.

A flabby sheaf is one for which this obstruction never appears. Every local section can be extended to a larger open set. This makes flabby sheaves very useful in cohomology, because cohomology measures obstructions to gluing and extending local data.

Flabby sheaves and exact sequences

Suppose

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is an exact sequence of sheaves.

In general, taking sections over an open set U gives a left exact sequence

$$0 \longrightarrow \mathcal{F}_1(U) \longrightarrow \mathcal{F}_2(U) \longrightarrow \mathcal{F}_3(U),$$

but the last map need not be surjective.

However, if $\mathcal{F}_1$ is flabby, then local liftings of sections of $\mathcal{F}_3$ can be modified and glued globally. Therefore the map

$$\mathcal{F}_2(U) \longrightarrow \mathcal{F}_3(U)$$

is surjective for every open set $U \subseteq X$.

Hence, if $\mathcal{F}_1$ is flabby, then

$$0 \longrightarrow \mathcal{F}_1(U) \longrightarrow \mathcal{F}_2(U) \longrightarrow \mathcal{F}_3(U) \longrightarrow 0$$

is exact for every open set U .

Quotients of flabby sheaves

Suppose again that

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is exact, and assume that $\mathcal{F}_1$ and $\mathcal{F}_2$ are flabby.

Then $\mathcal{F}_3$ is also flabby.

Indeed, let $V \subseteq U$ be open and take

$$s \in \mathcal{F}_3(V).$$

Since $\mathcal{F}_1$ is flabby, the map

$$\mathcal{F}_2(V) \longrightarrow \mathcal{F}_3(V)$$

is surjective. Therefore s lifts to some

$$t \in \mathcal{F}_2(V).$$

Since $\mathcal{F}_2$ is flabby, t extends to a section

$$\tilde{t} \in \mathcal{F}_2(U).$$

Its image in $\mathcal{F}_3(U)$ extends s . Hence

$$\mathcal{F}_3(U) \longrightarrow \mathcal{F}_3(V)$$

is surjective, so $\mathcal{F}_3$ is flabby.

Exactness of global sections for long exact sequences

Let

$$0 \longrightarrow \mathcal{F}_0 \longrightarrow \mathcal{F}_1 \longrightarrow \cdots \longrightarrow \mathcal{F}_{p-1} \longrightarrow \mathcal{F}_p \longrightarrow \mathcal{F}_{p+1} \longrightarrow \cdots$$

be a long exact sequence of flabby sheaves.

For each p , define

$$Z_p = \ker(\mathcal{F}_p \rightarrow \mathcal{F}_{p+1}).$$

Then there are short exact sequences

$$0 \longrightarrow Z_p \longrightarrow \mathcal{F}_p \longrightarrow Z_{p+1} \longrightarrow 0.$$

Using the fact that quotients of flabby sheaves by flabby subsheaves are flabby, one proves inductively that all the sheaves Z_p are flabby.

Then, since Z_p is flabby, the map

$$\mathcal{F}_p(U) \longrightarrow Z_{p+1}(U)$$

is surjective for every open set U .

But

$$Z_{p+1}(U) = \ker(\mathcal{F}_{p+1}(U) \rightarrow \mathcal{F}_{p+2}(U)).$$

Therefore

$$\operatorname{Im}(\mathcal{F}_p(U) \rightarrow \mathcal{F}_{p+1}(U)) = \ker(\mathcal{F}_{p+1}(U) \rightarrow \mathcal{F}_{p+2}(U)).$$

Hence the sequence

$$0 \longrightarrow \mathcal{F}_0(U) \longrightarrow \mathcal{F}_1(U) \longrightarrow \cdots \longrightarrow \mathcal{F}_p(U) \longrightarrow \mathcal{F}_{p+1}(U) \longrightarrow \cdots$$

is exact.

Thus long exact sequences of flabby sheaves remain exact after taking sections over any open set.

The Čech complex

Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open covering of X .

The Čech cochains of $\mathcal{F}$ with respect to $\mathcal{U}$ are

$$C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p \in I} \mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p}).$$

Thus

$$C^0(\mathcal{U}, \mathcal{F}) = \prod_i \mathcal{F}(U_i),$$

$$C^1(\mathcal{U}, \mathcal{F}) = \prod_{i,j} \mathcal{F}(U_i \cap U_j),$$

and

$$C^2(\mathcal{U}, \mathcal{F}) = \prod_{i,j,k} \mathcal{F}(U_i \cap U_j \cap U_k).$$

The Čech differential

$$\delta : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^{p+1}(\mathcal{U}, \mathcal{F})$$

is defined by

$$(\delta s)_{i_0, \dots, i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k s_{i_0, \dots, \widehat{i_k}, \dots, i_{p+1}} \Big|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}}.$$

For example, if $s = (s_i) \in C^0(\mathcal{U}, \mathcal{F})$, then

$$(\delta s)_{ij} = s_j|_{U_i \cap U_j} - s_i|_{U_i \cap U_j}.$$

Thus $\delta s = 0$ means that the local sections s_i agree on overlaps, so they glue to a global section.

For a 1-cochain (s_{ij}) , one has

$$(\delta s)_{ijk} = s_{jk}|_{U_i \cap U_j \cap U_k} - s_{ik}|_{U_i \cap U_j \cap U_k} + s_{ij}|_{U_i \cap U_j \cap U_k}.$$

The Čech cohomology groups are

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = \frac{\ker(\delta : C^p \rightarrow C^{p+1})}{\operatorname{im}(\delta : C^{p-1} \rightarrow C^p)}.$$

Thus Čech cohomology measures whether compatible data on intersections comes from data on smaller intersections.

The sheaf Čech resolution

Instead of only using groups of sections, we may define sheaves

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p \in I} \mathcal{F}|_{U_{i_0} \cap \dots \cap U_{i_p}}.$$

Then

$$C^p(\mathcal{U}, \mathcal{F}) = \Gamma(X, \mathcal{C}^p(\mathcal{U}, \mathcal{F})).$$

There is an augmented sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^2(\mathcal{U}, \mathcal{F}) \longrightarrow \dots$$

This sequence is exact.

The reason is local. Exactness of sheaves can be checked on stalks. Fix a point $x \in X$. Choose $i \in I$ such that $x \in U_i$.

On the stalk at x , the covering has a distinguished member U_i containing x . The resulting stalk complex is contractible. A contracting homotopy is given by inserting the fixed index i :

$$h(s)_{i_0, \dots, i_{p-1}} = s_{i, i_0, \dots, i_{p-1}}.$$

One checks that

$$\delta h + h\delta = \text{id}.$$

Therefore the stalk complex is exact at every point x , and hence the sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \cdots$$

is exact as a sequence of sheaves.

Why the sheaves $\mathcal{C}^p(\mathcal{U}, \mathcal{F})$ are flabby

Assume that $\mathcal{F}$ is flabby.

For every multi-index $(i_0, \dots, i_p)$, the sheaf

$$\mathcal{F}|_{U_{i_0} \cap \cdots \cap U_{i_p}}$$

is flabby. Indeed, if $V \subseteq W$ are open subsets of X , then the restriction map

$$\mathcal{F}(W \cap U_{i_0} \cap \cdots \cap U_{i_p}) \longrightarrow \mathcal{F}(V \cap U_{i_0} \cap \cdots \cap U_{i_p})$$

is surjective because $\mathcal{F}$ is flabby.

Also, a product of flabby sheaves is flabby, because restriction maps are taken component-wise, and products of surjective maps are surjective.

Therefore

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0, \dots, i_p} \mathcal{F}|_{U_{i_0} \cap \cdots \cap U_{i_p}}$$

is flabby for every $p \geq 0$.

Vanishing of higher Čech cohomology

We have an exact sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^2(\mathcal{U}, \mathcal{F}) \longrightarrow \cdots.$$

If $\mathcal{F}$ is flabby, then every sheaf

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F})$$

is also flabby.

Since global sections preserve exactness for long exact sequences of flabby sheaves, taking global sections gives an exact sequence

$$0 \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow C^0(\mathcal{U}, \mathcal{F}) \longrightarrow C^1(\mathcal{U}, \mathcal{F}) \longrightarrow C^2(\mathcal{U}, \mathcal{F}) \longrightarrow \cdots.$$

The cohomology of this complex in degree p is precisely

$$\check{H}^p(\mathcal{U}, \mathcal{F}).$$

Since the complex is exact in all positive degrees, we obtain

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \quad \text{for all } p > 0.$$

Thus

$\mathcal{F} \text{ flabby} \implies \check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \text{ for all } p > 0.$

2.1 Conceptual summary

The central idea is that flabby sheaves have no extension obstruction. Every local section over a smaller open set extends to every larger open set. Because of this, local lifting and gluing problems disappear.

Since higher Čech cohomology measures precisely the failure of local data on intersections to come from lower-degree data, flabby sheaves have no positive-degree Čech cohomology.

In short,

flabby sheaves are acyclic.

3 Examples of Flabby Sheaves and Vanishing of Čech Cohomology

3.1 Example 1: The Constant Sheaf is Usually Not Flabby

Let $X = \mathbb{R}$ and consider the constant sheaf $\mathbb{Z}_X$.

Take

$$U = \mathbb{R}, \quad V = (0, 1) \cup (2, 3).$$

A section of $\mathbb{Z}_X$ over V is a locally constant function. Since V has two connected components, we may define

$$s|_{(0,1)} = 1, \quad s|_{(2,3)} = 2.$$

Suppose s extended to a global section on $\mathbb{R}$. Since $\mathbb{R}$ is connected, every global section of $\mathbb{Z}_X$ must be constant.

However, a constant function cannot simultaneously take the values 1 and 2 on different components.

Therefore s cannot be extended to $\mathbb{R}$.

Hence the restriction map

$$\mathbb{Z}_X(\mathbb{R}) \longrightarrow \mathbb{Z}_X(V)$$

is not surjective.

Thus

$\mathbb{Z}_X$ is not flabby.

3.2 Example 2: A Skyscraper Sheaf is Flabby

Fix a point $x \in X$ and an abelian group A .

The skyscraper sheaf $i_{x,*}A$ is defined by

$$(i_{x,*}A)(U) = \begin{cases} A, & x \in U, \\ 0, & x \notin U. \end{cases}$$

Let $V \subseteq U$.

If $x \notin V$, then

$$(i_{x,*}A)(V) = 0,$$

so the restriction map is automatically surjective.

If $x \in V$, then also $x \in U$, and

$$(i_{x,*}A)(U) = A, \quad (i_{x,*}A)(V) = A.$$

The restriction map is simply

$$A \longrightarrow A,$$

which is the identity map.

Therefore every restriction map is surjective.

Hence

$$\boxed{i_{x,*}A \text{ is flabby.}}$$

Consequently,

$$\check{H}^p(\mathcal{U}, i_{x,*}A) = 0 \quad (p > 0).$$

3.3 Example 3: The Sheaf of All Functions

Let A be an abelian group and define

$$\mathcal{F}(U) = \{f : U \rightarrow A\}.$$

Take open sets

$$V \subseteq U.$$

Given a function

$$f : V \rightarrow A,$$

choose a fixed element $a_0 \in A$ and define

$$\tilde{f}(x) = \begin{cases} f(x), & x \in V, \\ a_0, & x \in U \setminus V. \end{cases}$$

Then

$$\tilde{f}|_V = f.$$

Thus every section extends.

Therefore

$$\boxed{\mathcal{F} \text{ is flabby.}}$$

3.4 Example 4: Continuous Functions are Not Flabby

Consider the sheaf

$$\mathcal{C}^0(U) = \{\text{continuous functions } U \rightarrow \mathbb{R}\}.$$

Let

$$U = \mathbb{R}, \quad V = (0, 1).$$

Define

$$f(x) = \frac{1}{x}.$$

The function is continuous on $(0, 1)$.

Suppose f extended to a continuous function

$$\tilde{f} : \mathbb{R} \rightarrow \mathbb{R}.$$

Near 0 we have

$$\frac{1}{x} \rightarrow +\infty.$$

Therefore no continuous extension exists.

Hence

$$\mathcal{C}^0(\mathbb{R}) \longrightarrow \mathcal{C}^0((0, 1))$$

is not surjective.

Thus

$\mathcal{C}^0$ is not flabby.

3.5 Example 5: Vanishing of Čech H^1 for a Flabby Sheaf

Suppose

$$X = U_1 \cup U_2$$

and $\mathcal{F}$ is flabby.

The Čech complex begins as

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{F}(U_1) \times \mathcal{F}(U_2) \longrightarrow \mathcal{F}(U_1 \cap U_2) \longrightarrow 0.$$

The differential is

$$\delta(s_1, s_2) = s_2|_{U_1 \cap U_2} - s_1|_{U_1 \cap U_2}.$$

Take

$$a \in \mathcal{F}(U_1 \cap U_2).$$

Since $\mathcal{F}$ is flabby, a extends to a section

$$s_2 \in \mathcal{F}(U_2).$$

Let

$$s_1 = 0.$$

Then

$$\delta(s_1, s_2) = a.$$

Therefore every 1-cocycle is a coboundary.

Hence

$$\check{H}^1(\mathcal{U}, \mathcal{F}) = 0.$$

3.6 Example 6: Understanding the Cocycle Condition

Let

$$X = U_1 \cup U_2 \cup U_3.$$

A Čech 1-cochain consists of sections

$$a_{12}, a_{13}, a_{23}.$$

These form a cocycle when

$$a_{23} - a_{13} + a_{12} = 0$$

on

$$U_1 \cap U_2 \cap U_3.$$

For an arbitrary sheaf, such overlap data may not come from sections on the individual open sets.

For a flabby sheaf, each section on an intersection can be extended to a larger open set.

Using these extensions one constructs sections

$$s_i \in \mathcal{F}(U_i)$$

such that

$$a_{ij} = s_j|_{U_i \cap U_j} - s_i|_{U_i \cap U_j}.$$

Hence

$$a = \delta(s).$$

Thus every cocycle is a coboundary.

This explains geometrically why higher Čech cohomology vanishes.

3.7 Example 7: Flabby Resolutions

The constant sheaf $\mathbb{Z}_X$ is generally not flabby.

However, it can be embedded into a flabby sheaf.

Let

$$\mathcal{G}^0(U) = \{f : U \rightarrow \mathbb{Z}\},$$

the sheaf of all integer-valued functions.

Then

$$\mathbb{Z}_X \hookrightarrow \mathcal{G}^0.$$

The quotient

$$\mathcal{G}^1 = \mathcal{G}^0 / \mathbb{Z}_X$$

is again used to continue the construction.

One obtains a flabby resolution

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{G}^0 \longrightarrow \mathcal{G}^1 \longrightarrow \mathcal{G}^2 \longrightarrow \cdots .$$

Since all $\mathcal{G}^i$ are flabby, one computes cohomology using global sections:

$$0 \longrightarrow \Gamma(X, \mathcal{G}^0) \longrightarrow \Gamma(X, \mathcal{G}^1) \longrightarrow \Gamma(X, \mathcal{G}^2) \longrightarrow \cdots .$$

This is one of the fundamental methods of sheaf cohomology.

3.8 Example 8: Computing Cohomology via a Flabby Resolution

Suppose $\mathcal{F}$ is any sheaf.

Choose a flabby resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \mathcal{I}^2 \longrightarrow \cdots .$$

Because each $\mathcal{I}^p$ is flabby,

$$H^q(X, \mathcal{I}^p) = 0 \quad (q > 0).$$

Therefore all cohomological information of $\mathcal{F}$ is contained in the complex of global sections

$$0 \longrightarrow \Gamma(X, \mathcal{I}^0) \longrightarrow \Gamma(X, \mathcal{I}^1) \longrightarrow \Gamma(X, \mathcal{I}^2) \longrightarrow \cdots .$$

The cohomology of this complex is

$$H^p(X, \mathcal{F}).$$

This illustrates why flabby sheaves play such a central role in modern algebraic geometry.

3.9 Summary Table

Sheaf	Flabby?	Reason
All functions $U \rightarrow A$	Yes	Arbitrary extensions exist
Skyscraper sheaf $i_{x,*}A$	Yes	Restrictions are identities or zero maps
Constant sheaf $\mathbb{Z}_X$	Usually No	Locally constant sections may not extend
Continuous functions $\mathcal{C}^0$	No	Continuous extension may fail
Holomorphic functions $\mathcal{O}_X$	No	Analytic continuation creates obstructions
Terms in a flabby resolution	Yes	Constructed to have surjective restrictions

3.10 Conceptual Conclusion

Flabby sheaves possess an extremely strong extension property:

$$\boxed{\text{Every local section extends to every larger open set.}}$$

Because local sections can always be extended, every Čech cocycle can be resolved into a coboundary.

Consequently,

$$\boxed{\check{H}^p(\mathcal{U}, \mathcal{F}) = 0 \quad (p > 0).}$$

This is the fundamental reason that flabby sheaves are called *acyclic sheaves* and are used throughout sheaf cohomology and algebraic geometry.

4 Problem 4: The Punctured Affine Plane is Not Affine

Let

$$X = \mathbb{A}_k^2 = \operatorname{Spec} k[x, y],$$

and let

$$U = X \setminus \{(x, y)\}$$

be the affine plane with the origin removed.

We cover U by the two standard affine open subsets

$$U = D(x) \cup D(y).$$

Indeed, a point of $\mathbb{A}_k^2$ is not the origin precisely when either $x \neq 0$ or $y \neq 0$.

Thus

$$D(x) = \operatorname{Spec} k[x, y]_x, \quad D(y) = \operatorname{Spec} k[x, y]_y,$$

and

$$D(x) \cap D(y) = D(xy) = \operatorname{Spec} k[x, y]_{xy}.$$

The Čech complex for $\mathcal{O}_U$ with respect to this cover is

$$0 \longrightarrow \mathcal{O}_U(U) \longrightarrow k[x, y]_x \oplus k[x, y]_y \longrightarrow k[x, y]_{xy} \longrightarrow H^1(U, \mathcal{O}_U) \longrightarrow 0.$$

The middle map is

$$(f, g) \longmapsto g - f.$$

Therefore

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, y]_{xy}}{k[x, y]_x + k[x, y]_y}.$$

Now

$$k[x, y]_{xy} = k[x, y, x^{-1}, y^{-1}].$$

Every element of this ring is a finite k -linear combination of monomials

$$x^i y^j, \quad i, j \in \mathbb{Z}.$$

The subring $k[x, y]_x$ contains all monomials

$$x^i y^j, \quad i \in \mathbb{Z}, j \geq 0.$$

Similarly, $k[x, y]_y$ contains all monomials

$$x^i y^j, \quad i \geq 0, j \in \mathbb{Z}.$$

Therefore, in the quotient

$$\frac{k[x, y, x^{-1}, y^{-1}]}{k[x, y, x^{-1}] + k[x, y, y^{-1}]},$$

all monomials disappear except those with both exponents negative.

Hence

$$H^1(U, \mathcal{O}_U) \cong \bigoplus_{i < 0, j < 0} kx^i y^j.$$

Thus $H^1(U, \mathcal{O}_U)$ has k -basis

$$\{x^i y^j \mid i < 0, j < 0\}.$$

In particular,

$$H^1(U, \mathcal{O}_U)$$

is infinite-dimensional over k .

If U were affine, then for the quasi-coherent sheaf $\mathcal{O}_U$, we would have

$$H^1(U, \mathcal{O}_U) = 0.$$

But we have just shown that

$$H^1(U, \mathcal{O}_U) \neq 0.$$

Therefore

$U = \mathbb{A}_k^2 \setminus \{0\} \text{ is not affine.}$

5 Problem 5: A Plane Curve and Its Čech Cohomology

Let $X \subseteq \mathbb{P}_k^2$ be the closed subscheme defined by one homogeneous polynomial

$$f(x_0, x_1, x_2) = 0$$

of degree d .

Assume

$$(1, 0, 0) \notin X.$$

We first show that X is covered by the two affine open subsets

$$U = X \cap D_+(x_1), \quad V = X \cap D_+(x_2).$$

The complement of

$$D_+(x_1) \cup D_+(x_2)$$

inside $\mathbb{P}_k^2$ is the single point

$$(1, 0, 0).$$

Since

$$(1, 0, 0) \notin X,$$

every point of X lies in either $D_+(x_1)$ or $D_+(x_2)$.

Hence

$$X = U \cup V.$$

Both U and V are affine.

On $D_+(x_1)$, set

$$t = \frac{x_0}{x_1}, \quad z = \frac{x_2}{x_1}.$$

Then

$$U \simeq \operatorname{Spec} k[t, z]/(f(t, 1, z)).$$

Similarly, on $D_+(x_2)$, set

$$s = \frac{x_0}{x_2}, \quad w = \frac{x_1}{x_2}.$$

Then

$$V \simeq \operatorname{Spec} k[s, w]/(f(s, w, 1)).$$

Their intersection is

$$U \cap V = X \cap D_+(x_1 x_2).$$

This is affine.

Since

$$(1, 0, 0) \notin X,$$

we have

$$f(1, 0, 0) \neq 0.$$

Therefore the coefficient of x_0^d in f is nonzero. After scaling, we may assume that this coefficient is 1.

On the intersection $U \cap V$, use coordinates

$$u = \frac{x_0}{x_1}, \quad v = \frac{x_2}{x_1}.$$

Since $x_2 \neq 0$ on $U \cap V$, the coordinate v is invertible.

Thus

$$\mathcal{O}_X(U \cap V) = k[u, v, v^{-1}]/(f(u, 1, v)).$$

Because $f(u, 1, v)$ is monic of degree d in u , every element of $\mathcal{O}_X(U \cap V)$ can be represented uniquely as

$$\sum_{r=0}^{d-1} a_r(v)u^r, \quad a_r(v) \in k[v, v^{-1}].$$

The Čech complex for the cover $X = U \cup V$ is

$$0 \longrightarrow \mathcal{O}_X(X) \longrightarrow \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \longrightarrow \mathcal{O}_X(U \cap V) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow 0.$$

Since U , V , and $U \cap V$ are affine, this complex computes H^0 and H^1 .

The global sections are the elements of

$$\mathcal{O}_X(U) \oplus \mathcal{O}_X(V)$$

which agree on $U \cap V$. For a projective plane curve, the only global regular functions are constants. Hence

$$H^0(X, \mathcal{O}_X) \cong k,$$

so

$$\boxed{\dim_k H^0(X, \mathcal{O}_X) = 1.}$$

Now we compute $H^1(X, \mathcal{O}_X)$.

The cokernel of

$$\mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \longrightarrow \mathcal{O}_X(U \cap V)$$

is represented by the principal parts on the overlap that cannot come from sections on U or on V .

A basis for the quotient is given by the monomials

$$u^r v^{-m}$$

with

$$1 \leq r, \quad 1 \leq m, \quad r + m \leq d - 1.$$

Equivalently,

$$1 \leq r \leq d - 2, \quad 1 \leq m \leq d - 1 - r.$$

For each $r = 1, \dots, d - 2$, the integer m can take

$$d - 1 - r$$

values.

Therefore the number of such monomials is

$$\sum_{r=1}^{d-2} (d - 1 - r) = (d - 2) + (d - 3) + \cdots + 1.$$

Hence

$$\dim_k H^1(X, \mathcal{O}_X) = 1 + 2 + \cdots + (d - 2).$$

Therefore

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.$$

Thus

$$\dim_k H^0(X, \mathcal{O}_X) = 1$$

and

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.$$

If X is nonsingular, then

$$\dim_k H^1(X, \mathcal{O}_X)$$

is the genus of the plane curve. Hence a nonsingular plane curve of degree d has genus

$$g = \frac{(d-1)(d-2)}{2}.$$

6 Problem 6: Relative Differentials

(a) Construction and Universal Property of $\Omega_{B/A}$

Let A be a ring, let B be an A -algebra, and let M be a B -module.

An A -derivation of B into M is a map

$$d : B \rightarrow M$$

such that:

$$d(b + b') = d(b) + d(b'),$$

$$d(bb') = b d(b') + b' d(b),$$

and

$$d(a) = 0 \quad \text{for all } a \in A.$$

Let

$$\mu : B \otimes_A B \longrightarrow B$$

be the multiplication map

$$\mu(b \otimes b') = bb'.$$

Let

$$I = \ker(\mu).$$

Define

$$\Omega_{B/A} = I/I^2.$$

For $b \in B$, define

$$db = 1 \otimes b - b \otimes 1.$$

Since

$$\mu(1 \otimes b - b \otimes 1) = b - b = 0,$$

we have

$$db \in I.$$

Thus db defines an element of I/I^2 .

We check that

$$d : B \rightarrow I/I^2$$

is an A -derivation.

First,

$$d(b + b') = 1 \otimes (b + b') - (b + b') \otimes 1.$$

Therefore

$$d(b + b') = db + db'.$$

Second,

$$d(bb') = 1 \otimes bb' - bb' \otimes 1.$$

Modulo I^2 , one has

$$d(bb') = b db' + b' db.$$

Indeed, this identity is exactly the image in I/I^2 of the standard identity expressing the Leibniz rule.

Finally, if $a \in A$, then

$$da = 1 \otimes a - a \otimes 1 = 0$$

because the tensor product is taken over A .

So d is an A -derivation.

Now let

$$d' : B \rightarrow M$$

be any A -derivation.

Define a map

$$\varphi : B \otimes_A B \rightarrow M$$

by

$$\varphi(b \otimes b') = b d'(b').$$

The derivation rule implies that φ sends I^2 to zero after restriction to I . Hence $\varphi|_I$ factors through I/I^2 .

Therefore there exists a B -linear map

$$f : I/I^2 \rightarrow M$$

such that

$$f(db) = d'(b).$$

Thus

$$d' = f \circ d.$$

The map f is unique because $\Omega_{B/A} = I/I^2$ is generated by the elements db .

Therefore $\Omega_{B/A}$, together with $d : B \rightarrow \Omega_{B/A}$, satisfies the universal property: for every A -derivation $d' : B \rightarrow M$, there exists a unique B -linear map

$$f : \Omega_{B/A} \rightarrow M$$

such that

$$d' = f \circ d.$$

Hence

$$\boxed{\Omega_{B/A} = I/I^2.}$$

(b) Computation of $\Omega_{k[x_1, \dots, x_n]/k}$

Let

$$A = k[x_1, \dots, x_n].$$

We claim that

$$\Omega_{A/k} \cong A dx_1 \oplus \cdots \oplus A dx_n.$$

Define

$$d : A \rightarrow \bigoplus_{i=1}^n A dx_i$$

by

$$d(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

This map is k -linear and satisfies

$$d(fg) = f dg + g df$$

because ordinary partial differentiation satisfies the product rule.

Also,

$$d(c) = 0 \quad (c \in k).$$

Therefore d is a k -derivation.

Now let

$$D : A \rightarrow M$$

be any k -derivation into an A -module M .

Since A is a polynomial ring, D is determined by the values

$$D(x_1), \dots, D(x_n).$$

Define an A -linear map

$$\varphi : A dx_1 \oplus \cdots \oplus A dx_n \longrightarrow M$$

by

$$\varphi(dx_i) = D(x_i).$$

Then for every polynomial f ,

$$\varphi(df) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} D(x_i).$$

By the usual formula for derivations on a polynomial ring,

$$D(f) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} D(x_i).$$

Thus

$$D = \varphi \circ d.$$

The map φ is unique because $dx_1, \dots, dx_n$ form a basis.

Hence

$$\Omega_{k[x_1, \dots, x_n]/k} \cong \bigoplus_{i=1}^n k[x_1, \dots, x_n] dx_i.$$

So $\Omega_{A/k}$ is the free A -module generated by

$$dx_1, \dots, dx_n.$$

Moreover, for every polynomial $f \in A$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

(c) The Sheaf of Differentials on Affine Schemes

Let

$$X = \operatorname{Spec} B, \quad Y = \operatorname{Spec} A,$$

where B is an A -algebra.

Then

$$X \times_Y X = \operatorname{Spec}(B \otimes_A B).$$

The diagonal morphism

$$\Delta : X \rightarrow X \times_Y X$$

corresponds to the multiplication map

$$B \otimes_A B \rightarrow B, \quad b \otimes b' \mapsto bb'.$$

Let

$$I = \ker(B \otimes_A B \rightarrow B).$$

The conormal sheaf of the diagonal closed immersion is

$$I/I^2$$

sheafified on X .

By part (a),

$$I/I^2 = \Omega_{B/A}.$$

Therefore

$$\Omega_{X/Y} \cong \widetilde{\Omega_{B/A}}.$$

Hence

$$\boxed{\Omega_{\mathrm{Spec} B / \mathrm{Spec} A} \cong \widetilde{\Omega_{B/A}}.}$$

(d) Differentials on Affine Space

Let

$$X = \mathbb{A}_k^n = \mathrm{Spec} k[x_1, \dots, x_n].$$

By part (b),

$$\Omega_{k[x_1, \dots, x_n]/k} \cong \bigoplus_{i=1}^n k[x_1, \dots, x_n] dx_i.$$

Sheafifying this module on $\mathbb{A}_k^n$, we obtain

$$\Omega_{\mathbb{A}_k^n / \mathrm{Spec} k} \cong \bigoplus_{i=1}^n \mathcal{O}_{\mathbb{A}_k^n} dx_i.$$

Therefore

$$\boxed{\Omega_{\mathbb{A}_k^n / \mathrm{Spec} k} \cong \mathcal{O}_{\mathbb{A}_k^n}^{\oplus n}.}$$

Thus $\Omega_{\mathbb{A}_k^n / \mathrm{Spec} k}$ is a free $\mathcal{O}_{\mathbb{A}_k^n}$ -module of rank n , generated by

$$dx_1, \dots, dx_n.$$

7 Theory: Affine Covers, Čech Cohomology, Plane Curves, and Relative Differentials

7.1 Affine Schemes and Vanishing of Higher Cohomology

A fundamental theorem in algebraic geometry says that if X is affine and $\mathcal{F}$ is quasi-coherent, then

$$H^p(X, \mathcal{F}) = 0 \quad \text{for all } p > 0.$$

In particular, for the structure sheaf,

$$H^1(X, \mathcal{O}_X) = 0$$

whenever X is affine.

Therefore, to prove that a scheme U is not affine, it is enough to find

$$H^1(U, \mathcal{O}_U) \neq 0.$$

For example, for the punctured affine plane

$$U = \mathbb{A}_k^2 \setminus \{0\},$$

we use the cover

$$U = D(x) \cup D(y).$$

The overlap is

$$D(x) \cap D(y) = D(xy).$$

The Čech computation gives

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, y]_{xy}}{k[x, y]_x + k[x, y]_y}.$$

Now

$$k[x, y]_{xy} = k[x, y, x^{-1}, y^{-1}].$$

The monomials which survive in the quotient are exactly those of the form

$$x^i y^j \quad (i < 0, j < 0).$$

Hence

$$H^1(U, \mathcal{O}_U) \cong \bigoplus_{i < 0, j < 0} kx^i y^j.$$

This is infinite-dimensional over k , and in particular nonzero. Therefore

$$\mathbb{A}_k^2 \setminus \{0\}$$

is not affine.

The principle is:

$$\boxed{H^1(U, \mathcal{O}_U) \neq 0 \implies U \text{ is not affine.}}$$

7.2 Čech Cohomology on a Two-Open Affine Cover

Suppose

$$X = U \cup V$$

where U , V , and $U \cap V$ are affine.

Then the Čech complex for $\mathcal{O}_X$ is

$$0 \longrightarrow \mathcal{O}_X(X) \longrightarrow \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \longrightarrow \mathcal{O}_X(U \cap V) \longrightarrow 0.$$

The map is

$$(s_U, s_V) \longmapsto s_V|_{U \cap V} - s_U|_{U \cap V}.$$

Thus

$$H^0(X, \mathcal{O}_X) = \ker(\mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U \cap V)),$$

and

$$H^1(X, \mathcal{O}_X) = \operatorname{coker}(\mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U \cap V)).$$

Therefore H^0 consists of pairs of regular functions on U and V that agree on the overlap, while H^1 measures regular functions on the overlap which cannot be written as a difference of functions coming from U and V .

This is the reason why quotients such as

$$\frac{k[x, y]_{xy}}{k[x, y]_x + k[x, y]_y}$$

naturally appear in Čech cohomology.

7.3 Projective Plane Curves and the Genus Formula

Let

$$X \subseteq \mathbb{P}_k^2$$

be a plane curve defined by one homogeneous polynomial

$$f(x_0, x_1, x_2) = 0$$

of degree d .

Assume

$$(1, 0, 0) \notin X.$$

Then X is covered by

$$U = X \cap D_+(x_1), \quad V = X \cap D_+(x_2).$$

Indeed, the complement of

$$D_+(x_1) \cup D_+(x_2)$$

in $\mathbb{P}_k^2$ is the single point

$$(1, 0, 0),$$

which does not belong to X .

Thus

$$X = U \cup V.$$

The sets U , V , and $U \cap V$ are affine. Therefore the two-open Čech complex computes $H^0(X, \mathcal{O}_X)$ and $H^1(X, \mathcal{O}_X)$.

For a projective plane curve, global regular functions are constants, so

$$H^0(X, \mathcal{O}_X) \cong k.$$

Hence

$$\dim_k H^0(X, \mathcal{O}_X) = 1.$$

The first cohomology group $H^1(X, \mathcal{O}_X)$ measures the failure of regular functions on the overlap $U \cap V$ to be represented by functions on U and V .

For a plane curve of degree d , the explicit Čech computation gives

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.$$

If X is nonsingular, then this dimension equals the genus:

$$g(X) = \dim_k H^1(X, \mathcal{O}_X).$$

Therefore a nonsingular plane curve of degree d has genus

$$g = \frac{(d-1)(d-2)}{2}.$$

For example:

$$d = 1 \quad \Rightarrow \quad g = 0,$$

$$d = 2 \quad \Rightarrow \quad g = 0,$$

$$d = 3 \quad \Rightarrow \quad g = 1,$$

$$d = 4 \quad \Rightarrow \quad g = 3.$$

Thus lines and conics have genus 0, smooth cubics have genus 1, and smooth quartics have genus 3.

7.4 Kähler Differentials

Let

$$A \rightarrow B$$

be a ring homomorphism.

The module of relative differentials

$$\Omega_{B/A}$$

is the universal object representing A -derivations from B .

An A -derivation into a B -module M is a map

$$D : B \rightarrow M$$

such that

$$\begin{aligned} D(b + b') &= D(b) + D(b'), \\ D(bb') &= bD(b') + b'D(b), \end{aligned}$$

and

$$D(a) = 0 \quad (a \in A).$$

The universal property of $\Omega_{B/A}$ says that there exists an A -derivation

$$d : B \rightarrow \Omega_{B/A}$$

such that every A -derivation

$$D : B \rightarrow M$$

factors uniquely through d .

That is, there is a unique B -linear map

$$\varphi : \Omega_{B/A} \rightarrow M$$

such that

$$D = \varphi \circ d.$$

Equivalently,

$$\mathrm{Der}_A(B, M) \cong \mathrm{Hom}_B(\Omega_{B/A}, M).$$

Thus $\Omega_{B/A}$ represents the functor of A -derivations.

7.5 Construction of Differentials Using the Diagonal

The module $\Omega_{B/A}$ can be constructed as

$$\Omega_{B/A} = I/I^2,$$

where

$$I = \ker(B \otimes_A B \rightarrow B),$$

and the map

$$B \otimes_A B \rightarrow B$$

is multiplication:

$$b \otimes b' \longmapsto bb'.$$

The universal differential is

$$db = 1 \otimes b - b \otimes 1.$$

Indeed,

$$1 \otimes b - b \otimes 1 \in I$$

because multiplication sends it to

$$b - b = 0.$$

The quotient I/I^2 keeps only first-order information. Products of elements of I are second-order infinitesimals, and these are killed in I/I^2 .

Geometrically, this construction comes from the diagonal morphism

$$\Delta : X \rightarrow X \times_Y X.$$

When

$$X = \operatorname{Spec} B, \quad Y = \operatorname{Spec} A,$$

we have

$$X \times_Y X = \operatorname{Spec}(B \otimes_A B).$$

The diagonal corresponds to the multiplication map

$$B \otimes_A B \rightarrow B.$$

The ideal I defines the diagonal, and

$$I/I^2$$

is the conormal module of the diagonal.

Thus differentials measure first-order infinitesimal displacement away from the diagonal.

7.6 Differentials on Affine Space

Let

$$A = k[x_1, \dots, x_n].$$

Then

$$\Omega_{A/k} \cong A dx_1 \oplus \dots \oplus A dx_n.$$

Thus $\Omega_{A/k}$ is a free A -module of rank n .

For every polynomial $f \in A$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

This is the algebraic analogue of the total differential from calculus.

After sheafifying on

$$\mathbb{A}_k^n = \operatorname{Spec} k[x_1, \dots, x_n],$$

we get

$$\Omega_{\mathbb{A}_k^n / \operatorname{Spec} k} \cong \mathcal{O}_{\mathbb{A}_k^n}^{\oplus n}.$$

Therefore

$$\Omega_{\mathbb{A}_k^n / \operatorname{Spec} k}$$

is a free $\mathcal{O}_{\mathbb{A}_k^n}$ -module of rank n , generated by

$$dx_1, \dots, dx_n.$$

7.7 Conceptual Summary

The problems in this group are linked by the same principle:

$$\boxed{\text{local algebra controls global geometry.}}$$

For the punctured affine plane, the localizations

$$k[x, y]_x, \quad k[x, y]_y, \quad k[x, y]_{xy}$$

detect non-affineness through the nonzero cohomology group

$$H^1(U, \mathcal{O}_U).$$

For plane curves, affine charts and their overlap compute

$$H^0(X, \mathcal{O}_X) \quad \text{and} \quad H^1(X, \mathcal{O}_X),$$

leading to the genus formula

$$g = \frac{(d-1)(d-2)}{2}.$$

For relative differentials, the algebra

$$B \otimes_A B$$

and the diagonal ideal

$$I = \ker(B \otimes_A B \rightarrow B)$$

encode infinitesimal geometry through

$$\Omega_{B/A} = I/I^2.$$

Thus these problems introduce three central tools:

$$\boxed{\check{\text{C}}\text{ech cohomology, affine covers, and Kähler differentials.}}$$

8 Examples: Affine Covers, Čech Cohomology, Plane Curves, and Differentials

8.1 Example 1: A Basic Čech Computation on Two Opens

Let

$$X = U \cup V.$$

For the structure sheaf, the Čech complex is

$$0 \longrightarrow \mathcal{O}_X(X) \longrightarrow \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \longrightarrow \mathcal{O}_X(U \cap V) \longrightarrow 0.$$

The middle map is

$$(f, g) \mapsto g|_{U \cap V} - f|_{U \cap V}.$$

Therefore

$$H^0(X, \mathcal{O}_X) = \{(f, g) \mid f|_{U \cap V} = g|_{U \cap V}\},$$

and

$$H^1(X, \mathcal{O}_X) = \frac{\mathcal{O}_X(U \cap V)}{\mathcal{O}_X(U) + \mathcal{O}_X(V)}.$$

Thus H^1 measures functions on the overlap which cannot be produced from functions on the two pieces.

8.2 Example 2: The Punctured Affine Line is Affine

Let

$$X = \mathbb{A}_k^1 = \operatorname{Spec} k[x], \quad U = X \setminus \{0\}.$$

Then

$$U = D(x).$$

But

$$D(x) = \operatorname{Spec} k[x]_x = \operatorname{Spec} k[x, x^{-1}].$$

Therefore

$$\boxed{\mathbb{A}_k^1 \setminus \{0\} \text{ is affine.}}$$

This contrasts with

$$\mathbb{A}_k^2 \setminus \{0\},$$

which is not affine.

The difference is that in dimension 1, removing the origin gives one principal open set. In dimension 2, removing the origin gives

$$D(x) \cup D(y),$$

and the overlap creates nonzero H^1 .

8.3 Example 3: The Punctured Affine Plane and Surviving Monomials

Let

$$U = \mathbb{A}_k^2 \setminus \{0\} = D(x) \cup D(y).$$

Then

$$H^1(U, \mathcal{O}_U) \cong \frac{k[x, y, x^{-1}, y^{-1}]}{k[x, y, x^{-1}] + k[x, y, y^{-1}]}.$$

We examine monomials

$$x^i y^j.$$

A monomial belongs to $k[x, y, x^{-1}]$ if

$$j \geq 0.$$

A monomial belongs to $k[x, y, y^{-1}]$ if

$$i \geq 0.$$

Therefore a monomial survives in the quotient only when

$$i < 0 \quad \text{and} \quad j < 0.$$

Thus examples of nonzero cohomology classes are

$$x^{-1}y^{-1}, \quad x^{-2}y^{-1}, \quad x^{-1}y^{-3}.$$

Hence

$$H^1(U, \mathcal{O}_U)$$

has basis

$$\{x^i y^j \mid i < 0, j < 0\}.$$

This gives infinitely many independent cohomology classes.

8.4 Example 4: A Degree One Plane Curve

Let $X \subseteq \mathbb{P}_k^2$ be a line.

Then $d = 1$. The genus formula gives

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2} = \frac{0 \cdot (-1)}{2} = 0.$$

So

$$H^1(X, \mathcal{O}_X) = 0.$$

Also,

$$H^0(X, \mathcal{O}_X) \cong k.$$

A line in $\mathbb{P}^2$ is isomorphic to $\mathbb{P}^1$, and $\mathbb{P}^1$ has genus 0.

Thus

A line has genus 0.

8.5 Example 5: A Degree Two Plane Curve

Let $X \subseteq \mathbb{P}_k^2$ be a smooth conic.

Then $d = 2$. The formula gives

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(2-1)(2-2)}{2} = 0.$$

So

$$H^1(X, \mathcal{O}_X) = 0.$$

A smooth conic is also isomorphic to $\mathbb{P}^1$, so it has genus 0.

Thus

A smooth conic has genus 0.

For example,

$$X = V(x_0x_2 - x_1^2) \subseteq \mathbb{P}^2$$

is a conic, and its genus is 0.

8.6 Example 6: A Degree Three Plane Curve

Let $X \subseteq \mathbb{P}_k^2$ be a smooth cubic.

Then $d = 3$. The formula gives

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(3-1)(3-2)}{2} = 1.$$

Thus

$$\dim_k H^1(X, \mathcal{O}_X) = 1.$$

Therefore the genus is

$$g = 1.$$

Hence

A smooth plane cubic is an elliptic curve.

For example,

$$X = V(y^2z - x^3 - axz^2 - bz^3) \subseteq \mathbb{P}^2$$

is a smooth cubic when the discriminant is nonzero.

8.7 Example 7: A Degree Four Plane Curve

Let $X \subseteq \mathbb{P}_k^2$ be a smooth quartic.

Then $d = 4$. Hence

$$\dim_k H^1(X, \mathcal{O}_X) = \frac{(4-1)(4-2)}{2} = \frac{3 \cdot 2}{2} = 3.$$

So a smooth quartic has genus

$$g = 3.$$

Thus

A smooth plane quartic has genus 3.

8.8 Example 8: A Simple Kähler Differential Computation

Let

$$A = k[x].$$

Then

$$\Omega_{A/k} = A dx.$$

For

$$f = x^5 + 3x^2,$$

we compute

$$df = d(x^5 + 3x^2).$$

Using the usual derivative rule,

$$df = (5x^4 + 6x) dx.$$

Thus, in algebraic geometry, the differential behaves like ordinary calculus:

$$\boxed{d(f) = f'(x) dx.}$$

8.9 Example 9: Kähler Differentials in Two Variables

Let

$$A = k[x, y].$$

Then

$$\Omega_{A/k} = A dx \oplus A dy.$$

For

$$f = x^3y^2 + y^5,$$

we have

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

Now

$$\frac{\partial f}{\partial x} = 3x^2y^2,$$

and

$$\frac{\partial f}{\partial y} = 2x^3y + 5y^4.$$

Therefore

$$\boxed{df = 3x^2y^2 dx + (2x^3y + 5y^4) dy.}$$

8.10 Example 10: Differentials Detect Singular Points

Let

$$X = V(f) \subseteq \mathbb{A}_k^2,$$

where

$$f = x^2 - y^3.$$

Then

$$\mathcal{O}_X = k[x, y]/(x^2 - y^3).$$

The module of differentials is

$$\Omega_{\mathcal{O}_X/k} = \frac{\mathcal{O}_X dx \oplus \mathcal{O}_X dy}{(df)}.$$

Compute

$$df = 2x dx - 3y^2 dy.$$

Thus

$$\Omega_{\mathcal{O}_X/k} = \frac{\mathcal{O}_X dx \oplus \mathcal{O}_X dy}{(2x dx - 3y^2 dy)}.$$

At the origin $(0, 0)$, the relation becomes

$$0 = 0.$$

So there is no linear relation between dx and dy at the origin.

The cotangent space has dimension 2, but the curve should have dimension 1.

This indicates that the origin is singular.

Thus differentials detect the cusp singularity of

$$x^2 = y^3.$$

8.11 Example 11: Smooth Hypersurface Differential Relation

Let

$$X = V(f) \subseteq \mathbb{A}_k^2$$

with

$$f = x^2 + y^2 - 1.$$

Then

$$df = 2x dx + 2y dy.$$

The module of differentials is

$$\Omega_{\mathcal{O}_X/k} = \frac{\mathcal{O}_X dx \oplus \mathcal{O}_X dy}{(2x dx + 2y dy)}.$$

At a point $(a, b) \in X$, not both partial derivatives vanish, because

$$(a, b) \neq (0, 0)$$

on $x^2 + y^2 = 1$.

Thus the relation

$$2a \, dx + 2b \, dy = 0$$

is nontrivial at every point of X .

This reflects that

$$x^2 + y^2 - 1 = 0$$

is smooth.

8.12 Example 12: The Diagonal and First-Order Differences

Let

$$B = k[x].$$

Then

$$B \otimes_k B \cong k[x_1, x_2].$$

The multiplication map

$$B \otimes_k B \rightarrow B$$

corresponds to

$$x_1 \mapsto x, \quad x_2 \mapsto x.$$

Its kernel is

$$I = (x_2 - x_1).$$

Therefore

$$\Omega_{B/k} = I/I^2.$$

The class of

$$x_2 - x_1$$

corresponds to

$$dx.$$

Hence

$$\Omega_{k[x]/k} \cong k[x] \, dx.$$

Geometrically, the diagonal in

$$\mathbb{A}^1 \times \mathbb{A}^1$$

is defined by

$$x_2 - x_1 = 0.$$

The quotient I/I^2 remembers the first-order displacement normal to the diagonal.

8.13 Summary Table

Object	Computation	Meaning
$\mathbb{A}^1 \setminus \{0\}$	$D(x) = \operatorname{Spec} k[x, x^{-1}]$	Affine
$\mathbb{A}^2 \setminus \{0\}$	$H^1 \neq 0$	Not affine
Line in $\mathbb{P}^2$	$g = 0$	Rational curve
Smooth conic	$g = 0$	Also rational
Smooth cubic	$g = 1$	Elliptic curve
Smooth quartic	$g = 3$	Genus 3 curve
$k[x_1, \dots, x_n]$	$\Omega = \bigoplus A dx_i$	Free differentials
$k[x, y]/(f)$	$\Omega = (A dx \oplus A dy)/(df)$	Detects smoothness

8.14 Final Intuition

These examples show three recurring ideas.

First,

cohomology detects non-affineness.

Second,

Čech cohomology computes global geometry from affine charts.

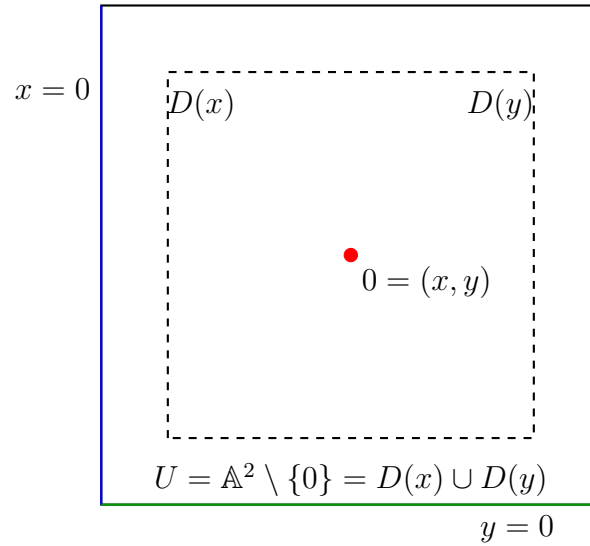
Third,

Kähler differentials encode infinitesimal geometry.

Therefore affine covers, Čech cohomology, and differentials are three different ways of turning local algebra into geometric information.

9 Charts, Tables, and Diagrams for Affine Covers, Cohomology, and Differentials

9.1 Diagram: The Punctured Affine Plane

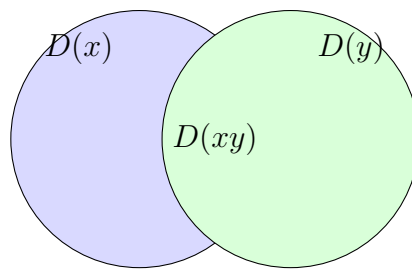


The removed point creates a global obstruction detected by

$$H^1(U, \mathcal{O}_U) \neq 0.$$

9.2 Diagram: The Čech Cover

$$U = D(x) \cup D(y)$$



The corresponding Čech complex is

$$0 \rightarrow \mathcal{O}(U) \rightarrow \mathcal{O}(D(x)) \oplus \mathcal{O}(D(y)) \rightarrow \mathcal{O}(D(xy)) \rightarrow H^1(U, \mathcal{O}) \rightarrow 0.$$

9.3 Table: Affine versus Non-Affine Examples

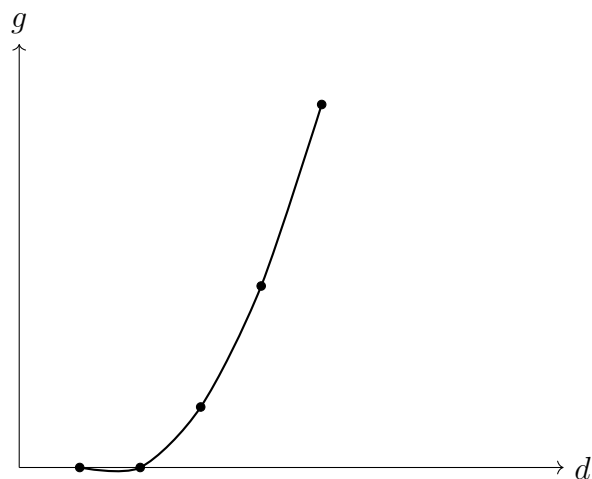
Space	Affine?	$H^1(\mathcal{O})$
$\mathbb{A}^1$	Yes	0
$\mathbb{A}^1 \setminus \{0\}$	Yes	0
$\mathbb{A}^2$	Yes	0
$\mathbb{A}^2 \setminus \{0\}$	No	$\neq 0$
$\mathbb{P}^1$	No	0
Smooth cubic curve	No	1
Smooth quartic curve	No	3

9.4 Chart: Genus of Smooth Plane Curves

For a smooth plane curve of degree d ,

$$g = \frac{(d-1)(d-2)}{2}.$$

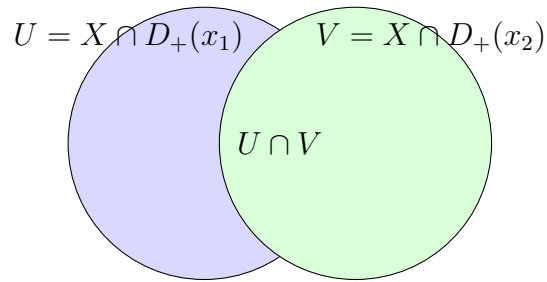
Degree d	Genus g
1	0
2	0
3	1
4	3
5	6
6	10
7	15
8	21



The genus grows quadratically with the degree.

9.5 Diagram: Projective Curve Covered by Two Affine Charts

$$X = (X \cap D_+(x_1)) \cup (X \cap D_+(x_2)).$$



The overlap computes

$$H^1(X, \mathcal{O}_X) = \frac{\mathcal{O}(U \cap V)}{\mathcal{O}(U) + \mathcal{O}(V)}.$$

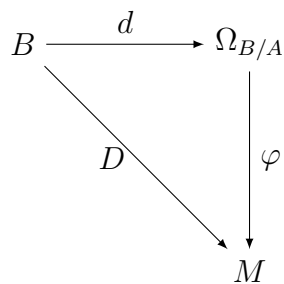
9.6 Table: Genus Classification of Plane Curves

Degree	Genus	Geometric Interpretation
1	0	Line
2	0	Conic
3	1	Elliptic curve
4	3	Quartic curve
5	6	Quintic curve

9.7 Diagram: Universal Property of Kähler Differentials

$$\Omega_{B/A}$$

represents all derivations.



Every derivation

$$D : B \rightarrow M$$

factors uniquely through

$$d : B \rightarrow \Omega_{B/A}.$$

9.8 Diagram: The Diagonal Construction

$$X \xrightarrow{\Delta} X \times_Y X$$

$$X \xrightarrow{\quad \Delta \quad} X \times_Y X$$

On rings:

$$B \otimes_A B \longrightarrow B.$$

The kernel is

$$I = \ker(B \otimes_A B \rightarrow B),$$

and

$$\Omega_{B/A} = I/I^2.$$

Thus differentials arise from the infinitesimal neighborhood of the diagonal.

9.9 Table: Differential Modules of Common Varieties

Variety	Module of Differentials
$\mathbb{A}^1$	$\mathcal{O} dx$
$\mathbb{A}^2$	$\mathcal{O} dx \oplus \mathcal{O} dy$
$\mathbb{A}^n$	$\mathcal{O}^{\oplus n}$
$V(f) \subseteq \mathbb{A}^2$	$\frac{\mathcal{O} dx \oplus \mathcal{O} dy}{(df)}$

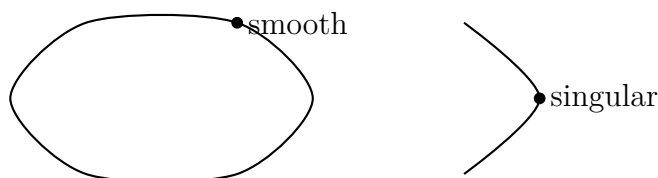
9.10 Diagram: Smooth versus Singular Point

For a hypersurface

$$f(x, y) = 0,$$

smoothness is determined by

$$df = f_x dx + f_y dy.$$



At a smooth point,

$$(df) \neq 0.$$

At a singular point,

$$df = 0.$$

This is why Kähler differentials detect singularities.

10 Explicit Čech Cohomology Computation for a Smooth Plane Curve

Let

$$X = V_+(f) \subset \mathbb{P}_k^2$$

be a projective plane curve defined by the homogeneous polynomial

$$f(x_0, x_1, x_2) = x_0^d + x_1^d + x_2^d,$$

where $d \geq 2$ and $\text{char}(k) \nmid d$.

Since

$$f(1, 0, 0) = 1 \neq 0,$$

the point $(1, 0, 0)$ does not belong to X . Therefore every point of X satisfies either $x_1 \neq 0$ or $x_2 \neq 0$. Hence

$$X = U \cup V,$$

where

$$U = X \cap D_+(x_1), \quad V = X \cap D_+(x_2).$$

We shall compute the Čech complex associated with this cover and determine $H^0(X, \mathcal{O}_X)$ and $H^1(X, \mathcal{O}_X)$.

10.1 The affine chart U

Inside $D_+(x_1)$ introduce affine coordinates

$$u = \frac{x_0}{x_1}, \quad v = \frac{x_2}{x_1}.$$

Substituting into the defining equation gives

$$f = x_1^d(u^d + 1 + v^d).$$

Therefore

$$U \simeq \operatorname{Spec} \left(k[u, v]/(u^d + v^d + 1) \right).$$

Define

$$A = k[u, v]/(u^d + v^d + 1).$$

The defining ideal is

$$I_U = (u^d + v^d + 1) \subset k[u, v].$$

Hence

$$U = \operatorname{Spec}(A)$$

is a closed affine subscheme of $\mathbb{A}^2$.

10.2 The affine chart V

Inside $D_+(x_2)$ introduce coordinates

$$s = \frac{x_0}{x_2}, \quad t = \frac{x_1}{x_2}.$$

Then

$$f = x_2^d(s^d + t^d + 1).$$

Hence

$$V \simeq \operatorname{Spec} \left(k[s, t]/(s^d + t^d + 1) \right).$$

Set

$$B = k[s, t]/(s^d + t^d + 1).$$

Its defining ideal is

$$I_V = (s^d + t^d + 1) \subset k[s, t].$$

Thus

$$V = \operatorname{Spec}(B).$$

10.3 The intersection $U \cap V$

The intersection consists of points where

$$x_1 \neq 0, \quad x_2 \neq 0.$$

In the U -coordinates this means

$$v = \frac{x_2}{x_1} \neq 0.$$

Therefore

$$U \cap V = D(v) \subset U.$$

Consequently

$$\mathcal{O}_X(U \cap V) = A_v = k[u, v, v^{-1}]/(u^d + v^d + 1).$$

The coordinate systems are related by

$$s = \frac{x_0}{x_2} = \frac{x_0/x_1}{x_2/x_1} = \frac{u}{v},$$

and

$$t = \frac{x_1}{x_2} = v^{-1}.$$

Hence

$$s = \frac{u}{v}, \quad t = v^{-1}.$$

10.4 The Čech complex

For the cover

$$X = U \cup V$$

the Čech complex of the structure sheaf is

$$0 \longrightarrow \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \xrightarrow{\delta} \mathcal{O}_X(U \cap V) \longrightarrow 0.$$

Substituting the coordinate rings gives

$$0 \longrightarrow A \oplus B \xrightarrow{\delta} A_v \longrightarrow 0,$$

where

$$\delta(a, b) = a - b.$$

Therefore

$$H^0(X, \mathcal{O}_X) = \ker(\delta),$$

and

$$H^1(X, \mathcal{O}_X) = \operatorname{coker}(\delta) = A_v / (A + B).$$

10.5 Computation of $H^0(X, \mathcal{O}_X)$

A regular function on a projective irreducible variety is constant.

Thus

$$H^0(X, \mathcal{O}_X) = k.$$

Hence

$\dim H^0(X, \mathcal{O}_X) = 1.$

10.6 Computation of $H^1(X, \mathcal{O}_X)$

Elements of A_v may contain negative powers of v .

The image of A contains all expressions with nonnegative powers of v .

The image of B eliminates many additional negative-power terms via the relations

$$s = \frac{u}{v}, \quad t = v^{-1}.$$

After quotienting by $A + B$, the surviving classes are represented by monomials

$$u^i v^{-j}$$

satisfying

$$i \geq 0, \quad j \geq 1, \quad i + j \leq d - 2.$$

Therefore a basis of $H^1(X, \mathcal{O}_X)$ is

$$\left\{ u^i v^{-j} ; i \geq 0, j \geq 1, i + j \leq d - 2 \right\}.$$

Counting these monomials gives

$$\dim H^1(X, \mathcal{O}_X) = \sum_{i=0}^{d-3} (d - 2 - i).$$

Hence

$$\dim H^1(X, \mathcal{O}_X) = 1 + 2 + \cdots + (d - 2).$$

Using

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

we obtain

$$\boxed{\dim H^1(X, \mathcal{O}_X) = \frac{(d-1)(d-2)}{2}.}$$

10.7 Example: a smooth cubic curve

Consider

$$X = V_+(x_0^3 + x_1^3 + x_2^3).$$

Here

$$d = 3.$$

Therefore

$$\dim H^1(X, \mathcal{O}_X) = \frac{(3-1)(3-2)}{2} = 1.$$

The condition

$$i + j \leq 1$$

gives only one basis element:

$$v^{-1}.$$

Hence

$$H^1(X, \mathcal{O}_X) = k \cdot v^{-1}.$$

Thus

$$\boxed{\dim H^1(X, \mathcal{O}_X) = 1.}$$

This agrees with the fact that a smooth cubic curve has genus 1.

10.8 Example: a smooth quartic curve

Consider

$$X = V_+(x_0^4 + x_1^4 + x_2^4).$$

Here

$$d = 4.$$

Hence

$$\dim H^1(X, \mathcal{O}_X) = \frac{(4-1)(4-2)}{2} = 3.$$

The condition

$$i + j \leq 2$$

gives the basis elements

$$v^{-1}, \quad v^{-2}, \quad uv^{-1}.$$

Therefore

$$H^1(X, \mathcal{O}_X) = \text{span}_k \{v^{-1}, v^{-2}, uv^{-1}\}.$$

Consequently

$$\boxed{\dim H^1(X, \mathcal{O}_X) = 3.}$$

This agrees with the genus formula for smooth plane quartics.

11 Kähler Differentials: Explicit Examples

Let A be a ring, B an A -algebra, and M a B -module. An A -derivation is a map

$$D : B \rightarrow M$$

such that

$$D(b + b') = D(b) + D(b'),$$

$$D(bb') = bD(b') + b'D(b),$$

and

$$D(a) = 0 \quad \text{for all } a \in A.$$

The module of relative differentials $\Omega_{B/A}$ is the universal object for such derivations.

11.1 Example 1: The affine parabola

Let

$$A = k, \quad B = k[x, y]/(y - x^2).$$

Geometrically,

$$X = \operatorname{Spec} B$$

is the affine parabola

$$y = x^2$$

inside

$$\mathbb{A}_k^2 = \operatorname{Spec} k[x, y].$$

The ideal defining this closed subscheme is

$$I = (y - x^2) \subset k[x, y].$$

Start with

$$R = k[x, y].$$

For affine space,

$$\Omega_{R/k} = R dx \oplus R dy.$$

Now pass to

$$B = R/I.$$

The relation

$$y - x^2 = 0$$

must also satisfy

$$d(y - x^2) = 0.$$

Compute:

$$d(y - x^2) = dy - 2x dx.$$

Therefore, in $\Omega_{B/k}$,

$$dy = 2x dx.$$

Hence

$$\Omega_{B/k} = \frac{B dx \oplus B dy}{(dy - 2x dx)}.$$

Since dy is determined by dx , we get

$$\Omega_{B/k} \cong B dx.$$

The universal derivation

$$d : B \rightarrow \Omega_{B/k}$$

is given by

$$d(x) = dx, \quad d(y) = dy = 2x dx.$$

For example,

$$d(x^3 + y) = 3x^2 dx + dy.$$

Using $dy = 2x dx$, this becomes

$$d(x^3 + y) = (3x^2 + 2x)dx.$$

Now let

$$M = B.$$

Define a k -derivation

$$D : B \rightarrow B$$

by

$$D(x) = 1.$$

Since $y = x^2$, we must have

$$D(y) = D(x^2) = 2x.$$

Thus

$$D(x) = 1, \quad D(y) = 2x.$$

The universal property says that there exists a unique B -module homomorphism

$$f : \Omega_{B/k} \rightarrow B$$

such that

$$D = f \circ d.$$

Since

$$\Omega_{B/k} \cong B dx,$$

the map f is determined by

$$f(dx) = 1.$$

Then

$$f(dy) = f(2x dx) = 2x.$$

So

$$D(x) = f(dx) = 1, \quad D(y) = f(dy) = 2x.$$

This confirms the universal property in this concrete case.

11.2 Example 2: The nodal curve

Let

$$A = k, \quad B = k[x, y]/(y^2 - x^2(x + 1)).$$

Geometrically,

$$X = \operatorname{Spec} B$$

is the affine curve

$$y^2 = x^2(x + 1)$$

inside

$$\mathbb{A}_k^2.$$

It is defined by the ideal

$$I = (y^2 - x^2(x + 1)) \subset k[x, y].$$

Let

$$R = k[x, y], \quad g = y^2 - x^2(x + 1).$$

Then

$$B = R/(g).$$

As before,

$$\Omega_{R/k} = R dx \oplus R dy.$$

Passing to the quotient means that the differential of the defining equation must vanish:

$$dg = 0.$$

Compute:

$$g = y^2 - x^3 - x^2.$$

Therefore

$$dg = 2y dy - (3x^2 + 2x) dx.$$

Hence in $\Omega_{B/k}$,

$$2y \, dy = (3x^2 + 2x) \, dx.$$

So

$$\Omega_{B/k} = \frac{B \, dx \oplus B \, dy}{(2y \, dy - (3x^2 + 2x) \, dx)}.$$

Equivalently,

$$\Omega_{B/k} = \frac{B \, dx \oplus B \, dy}{(-(3x^2 + 2x) \, dx + 2y \, dy)}.$$

The universal derivation is

$$d : B \rightarrow \Omega_{B/k},$$

with

$$d(x) = dx, \quad d(y) = dy.$$

For example,

$$d(xy) = x \, dy + y \, dx.$$

Also,

$$d(y^2) = 2y \, dy,$$

and

$$d(x^2(x+1)) = d(x^3 + x^2) = (3x^2 + 2x) \, dx.$$

Thus the relation

$$y^2 = x^2(x+1)$$

forces

$$2y \, dy = (3x^2 + 2x) \, dx.$$

Now let

$$M = B.$$

A k -derivation

$$D : B \rightarrow B$$

is determined by its values

$$D(x) = p, \quad D(y) = q,$$

where $p, q \in B$, but these values must respect the relation.

Applying D to

$$y^2 - x^2(x + 1) = 0$$

gives

$$D(y^2 - x^2(x + 1)) = 0.$$

Therefore

$$2yq - (3x^2 + 2x)p = 0.$$

So derivations $D : B \rightarrow B$ correspond to pairs

$$(p, q) \in B^2$$

satisfying

$$2yq = (3x^2 + 2x)p.$$

This is exactly the same relation as in the module of differentials.

The universal property says that each such derivation corresponds to a unique B -module map

$$f : \Omega_{B/k} \rightarrow B.$$

Indeed, define

$$f(dx) = p, \quad f(dy) = q.$$

This is well-defined precisely when the relation in $\Omega_{B/k}$ is preserved:

$$f(2y dy - (3x^2 + 2x) dx) = 0.$$

That condition is

$$2yq - (3x^2 + 2x)p = 0.$$

Thus

$$D = f \circ d.$$

11.3 Construction using I/I^2

Now we explain the abstract construction.

Let

$$\mu : B \otimes_A B \rightarrow B$$

be the multiplication map

$$\mu(b \otimes b') = bb'.$$

Let

$$I = \ker(\mu).$$

Then

$$\Omega_{B/A} = I/I^2.$$

The universal derivation is

$$d : B \rightarrow I/I^2$$

defined by

$$db = 1 \otimes b - b \otimes 1.$$

This satisfies

$$d(b + b') = db + db',$$

and

$$d(bb') = b db' + b' db.$$

Therefore it is an A -derivation.

For the parabola,

$$B = k[x, y]/(y - x^2).$$

We have

$$dx = 1 \otimes x - x \otimes 1,$$

$$dy = 1 \otimes y - y \otimes 1.$$

Because

$$y = x^2,$$

we get

$$dy = d(x^2) = 2x dx.$$

Thus

$$\Omega_{B/k} = \frac{B dx \oplus B dy}{(dy - 2x dx)}.$$

For the nodal curve,

$$B = k[x, y]/(y^2 - x^2(x + 1)).$$

The defining relation gives

$$d(y^2 - x^2(x + 1)) = 0.$$

Hence

$$2y dy - (3x^2 + 2x) dx = 0.$$

So

$$\Omega_{B/k} = \frac{B dx \oplus B dy}{(2y dy - (3x^2 + 2x) dx)}.$$

11.4 Geometric interpretation

For the parabola,

$$X = V(y - x^2) \subset \mathbb{A}^2,$$

the module of differentials is generated by one element:

$$\Omega_{B/k} \cong B dx.$$

This reflects the fact that the parabola is smooth and one-dimensional.

The relation

$$dy = 2x dx$$

means that the infinitesimal change of y is determined by the infinitesimal change of x .

For the nodal curve,

$$X = V(y^2 - x^2(x + 1)) \subset \mathbb{A}^2,$$

we have

$$\Omega_{B/k} = \frac{B dx \oplus B dy}{(2y dy - (3x^2 + 2x) dx)}.$$

At the point

$$(0, 0),$$

the coefficients of the relation vanish:

$$2y = 0, \quad 3x^2 + 2x = 0.$$

Thus the relation imposes no linear condition on dx and dy at the origin. This indicates that the tangent space at the origin is too large, which is the algebraic sign of a singular point.

Away from the singular point, the relation removes one tangent direction, so the curve behaves like a smooth one-dimensional curve.

12 Problem 8: Closed Immersion Criterion via Sections

Let k be an algebraically closed field, let

$$X \subseteq \mathbb{P}_k^m$$

be a closed subscheme, and let

$$\varphi : X \rightarrow \mathbb{P}_k^n$$

be a morphism over $\text{Spec } k$ induced by a line bundle $\mathcal{L}$ and global sections

$$s_0, \dots, s_n \in \Gamma(X, \mathcal{L}).$$

Let

$$V \subseteq \Gamma(X, \mathcal{L})$$

be the k -vector subspace spanned by these sections.

Assume that $\mathcal{L}$ is generated by the sections $s_0, \dots, s_n$, so that φ is defined everywhere.

We also assume:

1. Elements of V separate points. That is, for any two distinct closed points $P, Q \in X$, there exists $s \in V$ such that

$$s_P \in \mathfrak{m}_P \mathcal{L}_P$$

but

$$s_Q \notin \mathfrak{m}_Q \mathcal{L}_Q,$$

or vice versa.

2. Elements of V separate tangent vectors. That is, for every closed point $P \in X$, the set

$$\{s \in V \mid s_P \in \mathfrak{m}_P \mathcal{L}_P\}$$

spans the k -vector space

$$\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P.$$

We prove that

$$\varphi : X \rightarrow \mathbb{P}_k^n$$

is a closed immersion.

Step 1: Local form of the morphism

Since the sections $s_0, \dots, s_n$ generate $\mathcal{L}$, for every point $P \in X$ there is at least one index i such that

$$s_i(P) \neq 0.$$

Let

$$X_i = \{P \in X \mid s_i(P) \neq 0\}.$$

On X_i , the section s_i trivializes $\mathcal{L}$. Therefore each ratio

$$\frac{s_j}{s_i}$$

is a regular function on X_i .

On this open set, the morphism φ is given by

$$P \mapsto \left[\frac{s_0}{s_i}(P) : \dots : 1 : \dots : \frac{s_n}{s_i}(P) \right].$$

Thus locally the morphism is described by the regular functions

$$\frac{s_j}{s_i}.$$

These functions are precisely the pullbacks of affine coordinates on the standard chart

$$D_+(y_i) \subseteq \mathbb{P}_k^n.$$

Step 2: Point separation implies injectivity

We show that condition (a) implies that φ is injective on closed points.

Let $P, Q \in X$ be distinct closed points.

By condition (a), there exists $s \in V$ such that, after possibly swapping P and Q ,

$$s_P \in \mathfrak{m}_P \mathcal{L}_P$$

and

$$s_Q \notin \mathfrak{m}_Q \mathcal{L}_Q.$$

This means that

$$s(P) = 0$$

but

$$s(Q) \neq 0.$$

Suppose, for contradiction, that

$$\varphi(P) = \varphi(Q).$$

Since φ is defined by the projective coordinates

$$[s_0 : \cdots : s_n],$$

the values of all sections in V at P and Q must have the same projective vanishing behavior.

In particular, if a section in V vanishes at P , then the corresponding linear form vanishes at $\varphi(P)$, hence also at $\varphi(Q)$, and so the section vanishes at Q .

This contradicts the choice of s .

Therefore

$$\varphi(P) \neq \varphi(Q).$$

Hence φ is injective on closed points.

Step 3: The image is closed

Since

$$X \subseteq \mathbb{P}_k^m$$

is a closed subscheme of projective space, X is projective over k . In particular, X is proper over k .

The target

$$\mathbb{P}_k^n$$

is separated over k .

A morphism from a proper scheme to a separated scheme has closed image. Equivalently, using the fact allowed in the hint, since X is projective, the image

$$\varphi(X)$$

is closed in $\mathbb{P}_k^n$.

Let

$$Y = \varphi(X)$$

with its scheme-theoretic image structure.

Then φ factors as

$$X \xrightarrow{\psi} Y \hookrightarrow \mathbb{P}_k^n.$$

To prove that φ is a closed immersion, it is enough to show that

$$\psi : X \rightarrow Y$$

is an isomorphism.

Step 4: Reduction to the structure sheaf

The map $\psi : X \rightarrow Y$ is surjective by definition of Y .

By Step 2, it is injective on closed points.

Since ψ is proper and quasi-finite on closed points, and since we are allowed to use the hint that condition (a) implies that φ is a closed map onto its image, we obtain that ψ is a homeomorphism on underlying topological spaces.

Thus it remains to show that the morphism of structure sheaves

$$\mathcal{O}_Y \longrightarrow \psi_* \mathcal{O}_X$$

is an isomorphism.

Since ψ is already a homeomorphism, it is enough to prove that for every closed point $P \in X$, with

$$Q = \psi(P) \in Y,$$

the induced local ring map

$$\mathcal{O}_{Y,Q} \longrightarrow \mathcal{O}_{X,P}$$

is surjective.

Condition (b) is precisely what gives this surjectivity.

Step 5: Tangent separation and local coordinates

Fix a closed point

$$P \in X.$$

Choose an index i such that

$$s_i(P) \neq 0.$$

Then s_i trivializes $\mathcal{L}$ near P .

For every section $s \in V$, the quotient

$$\frac{s}{s_i}$$

is a regular function near P .

Moreover,

$$s_P \in \mathfrak{m}_P \mathcal{L}_P$$

if and only if

$$\frac{s}{s_i} \in \mathfrak{m}_P.$$

Since $s_i(P) \neq 0$, multiplication by s_i^{-1} gives an isomorphism

$$\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P \cong \mathfrak{m}_P / \mathfrak{m}_P^2.$$

Condition (b) therefore says that the classes of the functions

$$\frac{s}{s_i}, \quad s \in V, \quad s(P) = 0,$$

span the cotangent space

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

But the functions

$$\frac{s_j}{s_i}$$

are exactly the pullbacks of affine coordinate functions from the chart

$$D_+(y_i) \subseteq \mathbb{P}_k^n.$$

Hence the image of the local ring

$$\mathcal{O}_{Y,Q}$$

inside

$$\mathcal{O}_{X,P}$$

contains enough functions to generate

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

Equivalently, the induced map on cotangent spaces

$$\mathfrak{m}_Q / \mathfrak{m}_Q^2 \longrightarrow \mathfrak{m}_P / \mathfrak{m}_P^2$$

is surjective.

Dually, the tangent map

$$T_P X \longrightarrow T_Q Y$$

is injective.

Thus condition (b) says that φ does not collapse infinitesimal directions.

Step 6: Surjectivity of the local ring maps

Let

$$A = \mathcal{O}_{Y,Q}, \quad B = \mathcal{O}_{X,P}.$$

We have a local homomorphism

$$A \longrightarrow B.$$

Its image contains constants and maps the maximal ideal of A into the maximal ideal of B .

From Step 5, the induced map

$$\mathfrak{m}_A/\mathfrak{m}_A^2 \longrightarrow \mathfrak{m}_B/\mathfrak{m}_B^2$$

is surjective.

Equivalently, the image of $\mathfrak{m}_A$ generates

$$\mathfrak{m}_B$$

modulo $\mathfrak{m}_B^2$.

By Nakayama's lemma, the image of $\mathfrak{m}_A$ generates $\mathfrak{m}_B$ as a B -module.

Since the local functions coming from Y contain these generators of $\mathfrak{m}_B$, and since constants also come from Y , the local map

$$\mathcal{O}_{Y,Q} \longrightarrow \mathcal{O}_{X,P}$$

is surjective.

Therefore

$$\mathcal{O}_Y \longrightarrow \psi_* \mathcal{O}_X$$

is surjective.

Because ψ is already a homeomorphism onto Y , this proves that

$$\psi : X \rightarrow Y$$

is a closed immersion whose underlying topological map is onto Y .

Hence ψ is an isomorphism onto Y .

Step 7: Conclusion

We have factored φ as

$$X \xrightarrow{\psi} Y \hookrightarrow \mathbb{P}_k^n,$$

where

$$Y = \varphi(X)$$

is a closed subscheme of $\mathbb{P}_k^n$.

We have shown that

$$\psi : X \rightarrow Y$$

is an isomorphism.

Therefore

$$\varphi : X \rightarrow \mathbb{P}_k^n$$

identifies X with the closed subscheme $Y \subseteq \mathbb{P}_k^n$.

Hence

$$\boxed{\varphi : X \rightarrow \mathbb{P}_k^n \text{ is a closed immersion.}}$$

Conceptual Summary

The two hypotheses have clear geometric meanings.

Condition (a) says that the sections separate points:

$$P \neq Q \implies \varphi(P) \neq \varphi(Q).$$

Condition (b) says that the sections separate tangent directions:

$$\text{no nonzero tangent vector is collapsed by } \varphi.$$

Together, these imply that φ embeds X into projective space with the correct scheme structure.

Thus

$$\boxed{\text{point separation} + \text{tangent separation} \implies \text{closed immersion.}}$$

13 Concrete Examples: Closed Immersion Criterion via Sections

13.1 Example 1: The Identity Embedding of $\mathbb{P}^1$

Let

$$X = \mathbb{P}_k^1$$

with homogeneous coordinates

$$[u : v].$$

Take the line bundle

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(1).$$

The global sections of $\mathcal{O}_{\mathbb{P}^1}(1)$ are spanned by

$$u, v.$$

Thus

$$V = \langle u, v \rangle.$$

These sections define the morphism

$$\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$$

given by

$$[u : v] \longmapsto [u : v].$$

Thus φ is the identity morphism.

Step 1: The sections generate $\mathcal{L}$

At any point

$$[u : v] \in \mathbb{P}^1,$$

we cannot have both $u = 0$ and $v = 0$.

Therefore at least one of the two sections

$$u, v$$

is nonzero at every point.

Hence u and v generate

$$\mathcal{O}_{\mathbb{P}^1}(1).$$

Therefore φ is defined everywhere.

Step 2: The sections separate points

Let

$$P = [a : b], \quad Q = [c : d]$$

be two distinct points of $\mathbb{P}^1$.

Since $P \neq Q$, the vectors

$$(a, b) \quad \text{and} \quad (c, d)$$

are not scalar multiples.

Consider the section

$$s = bu - av.$$

Then

$$s(P) = ba - ab = 0.$$

Now evaluate at Q :

$$s(Q) = bc - ad.$$

If

$$s(Q) = 0,$$

then

$$bc - ad = 0,$$

which implies

$$[a : b] = [c : d].$$

This contradicts the assumption $P \neq Q$.

Therefore

$$s(P) = 0, \quad s(Q) \neq 0.$$

So the sections in $V = \langle u, v \rangle$ separate points.

Step 3: The sections separate tangent vectors

Take the point

$$P = [1 : 0].$$

On the affine chart $u \neq 0$, define the coordinate

$$t = \frac{v}{u}.$$

Then P corresponds to

$$t = 0.$$

The local ring is

$$\mathcal{O}_{\mathbb{P}^1, P} = k[t]_{(t)}.$$

Its maximal ideal is

$$\mathfrak{m}_P = (t).$$

Thus

$$\mathfrak{m}_P / \mathfrak{m}_P^2 = (t) / (t^2).$$

This is a one-dimensional k -vector space generated by the class of t .

Now we look at sections in V which vanish at P . Since

$$u(P) \neq 0, \quad v(P) = 0,$$

the section v vanishes at P .

Trivializing $\mathcal{O}_{\mathbb{P}^1}(1)$ by the nonvanishing section u , we get

$$\frac{v}{u} = t.$$

Modulo (t^2) , the class of t generates

$$\mathfrak{m}_P / \mathfrak{m}_P^2.$$

Therefore the vanishing sections span the cotangent space at P .

The same argument works at every point of $\mathbb{P}^1$, after choosing a chart where one of u or v is nonzero.

Hence the sections separate tangent vectors.

Conclusion

The sections u, v generate $\mathcal{O}_{\mathbb{P}^1}(1)$, separate points, and separate tangent vectors. Therefore, by the closed immersion criterion,

$$\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$$

is a closed immersion.

Indeed,

$$\boxed{\varphi([u : v]) = [u : v]}$$

is simply the identity morphism.

13.2 Example 2: The Degree Two Veronese Embedding

Now take again

$$X = \mathbb{P}_k^1,$$

but use the line bundle

$$\mathcal{L} = \mathcal{O}_{\mathbb{P}^1}(2).$$

The global sections of $\mathcal{O}_{\mathbb{P}^1}(2)$ are spanned by

$$u^2, uv, v^2.$$

Thus

$$V = \langle u^2, uv, v^2 \rangle.$$

These sections define the morphism

$$\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^2$$

given by

$$[u : v] \longmapsto [u^2 : uv : v^2].$$

This is the degree two Veronese embedding.

Let homogeneous coordinates on $\mathbb{P}^2$ be

$$[X : Y : Z].$$

Then under φ ,

$$X = u^2, \quad Y = uv, \quad Z = v^2.$$

Therefore

$$XZ - Y^2 = u^2v^2 - (uv)^2 = 0.$$

So the image lies in the conic

$$V(XZ - Y^2) \subseteq \mathbb{P}^2.$$

Step 1: The sections generate $\mathcal{O}_{\mathbb{P}^1}(2)$

At a point

$$[u : v] \in \mathbb{P}^1,$$

not both u and v are zero.

Therefore not all of

$$u^2, uv, v^2$$

vanish at the point.

Hence these sections generate

$$\mathcal{O}_{\mathbb{P}^1}(2).$$

Thus φ is defined everywhere.

Step 2: Point separation

Let

$$P = [a : b], \quad Q = [c : d]$$

be two points of $\mathbb{P}^1$.

Their images are

$$\varphi(P) = [a^2 : ab : b^2],$$

and

$$\varphi(Q) = [c^2 : cd : d^2].$$

Suppose

$$\varphi(P) = \varphi(Q).$$

Then

$$[a^2 : ab : b^2] = [c^2 : cd : d^2].$$

We show that this implies

$$[a : b] = [c : d].$$

If $a \neq 0$, then on the chart $X \neq 0$,

$$\frac{Y}{X} = \frac{ab}{a^2} = \frac{b}{a}.$$

Thus the image determines

$$\frac{b}{a}.$$

Similarly, if $c \neq 0$, the equality of images gives

$$\frac{b}{a} = \frac{d}{c}.$$

Hence

$$[a : b] = [c : d].$$

If $a = 0$, then $P = [0 : 1]$, and

$$\varphi(P) = [0 : 0 : 1].$$

Equality

$$\varphi(Q) = [0 : 0 : 1]$$

forces

$$c = 0,$$

so

$$Q = [0 : 1] = P.$$

Thus in all cases,

$$\varphi(P) = \varphi(Q) \implies P = Q.$$

Therefore φ is injective on points, and the sections separate points.

Step 3: Tangent separation at $P = [1 : 0]$

Take

$$P = [1 : 0].$$

On the chart $u \neq 0$, use the coordinate

$$t = \frac{v}{u}.$$

Then

$$[u : v] = [1 : t].$$

Under φ ,

$$[1 : t] \mapsto [1 : t : t^2].$$

Now use the affine chart $X \neq 0$ in $\mathbb{P}^2$, with affine coordinates

$$A = \frac{Y}{X}, \quad B = \frac{Z}{X}.$$

Then the morphism becomes

$$A = t, \quad B = t^2.$$

The local ring of $\mathbb{P}^1$ at P is

$$k[t]_{(t)}.$$

Its maximal ideal is

$$(t).$$

The cotangent space is

$$\mathfrak{m}_P / \mathfrak{m}_P^2 = (t) / (t^2).$$

Modulo (t^2) , we have

$$A = t, \quad B = t^2 \equiv 0.$$

Thus the pullback of the affine coordinate A gives the generator t of

$$\mathfrak{m}_P/\mathfrak{m}_P^2.$$

Therefore the induced map on cotangent spaces is surjective.

Equivalently, the tangent map is injective.

So the morphism separates tangent vectors at $P = [1 : 0]$.

Step 4: Tangent separation at an arbitrary point

The same argument works at every point.

On any affine chart of $\mathbb{P}^1$, after choosing a local coordinate t , the morphism has the local form

$$t \longmapsto (t, t^2)$$

up to a change of coordinates.

The important feature is that the first coordinate has linear term t .

Therefore, modulo t^2 , the pullback of a coordinate function from $\mathbb{P}^2$ gives the generator of

$$\mathfrak{m}_P/\mathfrak{m}_P^2.$$

Hence the induced cotangent map is surjective at every point.

So the sections

$$u^2, uv, v^2$$

separate tangent vectors everywhere.

Step 5: The image is the conic

We already showed that the image is contained in

$$V(XZ - Y^2).$$

Conversely, let

$$[X : Y : Z] \in V(XZ - Y^2) \subseteq \mathbb{P}^2.$$

If $X \neq 0$, then write

$$t = \frac{Y}{X}.$$

Since

$$XZ = Y^2,$$

we have

$$\frac{Z}{X} = \left(\frac{Y}{X}\right)^2 = t^2.$$

Thus

$$[X : Y : Z] = [1 : t : t^2] = \varphi([1 : t]).$$

If $X = 0$, then the equation

$$XZ - Y^2 = 0$$

gives

$$Y = 0.$$

So the point is

$$[0 : 0 : 1] = \varphi([0 : 1]).$$

Therefore

$$\varphi(\mathbb{P}^1) = V(XZ - Y^2).$$

Conclusion

The sections

$$u^2, uv, v^2$$

generate $\mathcal{O}_{\mathbb{P}^1}(2)$, separate points, and separate tangent vectors.

Therefore the morphism

$$\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^2, \quad [u : v] \mapsto [u^2 : uv : v^2],$$

is a closed immersion.

Its image is the closed conic

$$XZ - Y^2 = 0.$$

Thus

$$\boxed{\mathbb{P}^1 \cong V(XZ - Y^2) \subseteq \mathbb{P}^2.}$$

This gives a concrete example of the closed immersion criterion.